A Sea of Lies: GPS Spoofing and the Emerging Maritime Cyber Battlefield

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Abstract—Recent tensions around the Strait of Hormuz highlight a growing dimension of maritime conflict: cyber attacks on navigation systems, including location spoofing, identification manipulation, and electronic interference. These incidents underscore systemic vulnerabilities in maritime infrastructure. This article examines emerging threats, connects recent events to longstanding weaknesses, and outlines security challenges.

The recent escalation of conflict around the Strait of Hormuz, culminating in disruptions and closures of one of the world's most critical maritime chokepoints, has underscored a fundamental shift in the nature of modern conflict [1], [2]. With around 90% of global trade transported by sea, disruptions to maritime operations now carry immediate and far-reaching economic consequences [3]. Increasingly, these disruptions are not solely kinetic. Instead, they are enabled and amplified by cyber operations that target the digital infrastructure underlying maritime navigation and logistics. Maritime cybersecurity is no longer a peripheral concern; it is becoming a centerpiece of modern warfare.

This shift is particularly evident in the growing prevalence of cyber attacks on navigation systems, including the Automated Identification System (AIS), Global Navigation Satellite Systems (GNSS), and supporting onboard and shore-based infrastructure. AIS, originally designed for safety and collision avoidance through open broadcast of vessel identity, position, and intent, has become a critical component of maritime situational awareness [4]. However, its lack of authentication and inherent trust assumptions expose it to a wide range of attacks, including location spoofing, identity spoofing, message injection, and deliberate disablement. These vulnerabilities are not merely theoretical. Mariners report direct encounters with AIS and GPS spoofing, jamming, and other forms of interference affecting navigation equipment in operational settings [5].

Recent large-scale empirical evidence further demonstrates that these threats are not isolated incidents but part of a broader, structured phenomenon. Analysis of global AIS data reveals tens of thousands of spoofing events affecting thousands of vessels, with persistent hotspots concentrated in conflict-adjacent regions such as the Black Sea, Eastern Mediterranean, and the Strait of Hormuz [6]. These events exhibit distinct spatial patterns, including circular "ghost ship" trajectories, linear displacements, and coordinated convergence of multiple vessels to fictitious locations, indicating deliberate and often large-scale interference. Critically, such spoofing activity has already led to real-world consequences, including vessel groundings, collisions, and operational disruptions, as exemplified by incidents like the MSC Antonia grounding in the Red

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Sea [5].

At the same time, the maritime environment presents unique challenges that amplify these risks. Ships operate as complex cyber-physical systems with tightly coupled IT and operational technology, often relying on legacy infrastructure, heterogeneous equipment, and limited onboard cybersecurity expertise. Mariners themselves report gaps in training, detection, and response capabilities, as well as increasing exposure to cyber threats ranging from ransomware to navigation manipulation [5]. The result is a domain where cyber attacks can directly translate into physical and economic consequences at global scale.

In this article, we provide a focused overview of AIS as a foundational maritime system, examine its core vulnerabilities, and discuss how these weaknesses are being actively exploited in modern conflicts. We survey key attack classes across three layers: signal manipulation, data injection, and software exploitation, and place them in the broader context of cyber-physical maritime threats. We then examine how AIS spoofing has become a deliberate instrument of modern conflict, with persistent hotspots concentrated in geopolitically sensitive regions, and real-world consequences exemplified by recent incidents in the Strait of Hormuz [7].

Finally, we outline the key open challenges in securing maritime navigation systems. These include the lack of robust signal authentication mechanisms in civilian GNSS, the difficulty of validating unauthenticated AIS data at scale, and the challenge of detecting spoofing without reliable ground truth in a high-volume, globally distributed environment. These challenges are compounded by legacy and heterogeneous maritime systems, as well as vulnerabilities in AIS software itself, including insecure parsing and insufficient input validation that can allow malformed or adversarial messages to propagate through navigation pipelines. Addressing these issues will require advances in multi-modal sensing, large-scale anomaly detection, and cross-domain data validation, alongside stronger software hardening practices for AIS implementations. Progress will also depend on closer collaboration across academia, industry, and government. As maritime systems continue to digitize, securing them will be essential not only for safety and efficiency, but for maintaining stability in an increasingly contested global environment.

Automatic Identification System (AIS): Architecture and the Trust Problem

Modern maritime navigation depends on a continuous exchange of position and identity data between vessels, port authorities, and monitoring systems. At the center of this exchange is the Automatic Identification System (AIS), a VHF radio-based protocol mandated by the International Maritime Organization for vessels over 300 gross tons [8]. Understanding how AIS works – and why it was built the way it was – is essential to understanding why it has become such an attractive target. AIS operates as a continuous broadcast system. Each vessel's transponder receives positioning data from an onboard GPS receiver, packages it with static vessel information such as name, flag, and Maritime Mobile Service Identity (MMSI) number, and broadcasts it over open VHF channels at regular intervals. Nearby vessels, coastal receivers, and satellite-based listeners ingest these broadcasts to build a shared picture of maritime traffic. Figure 1 illustrates this end-to-end pipeline.

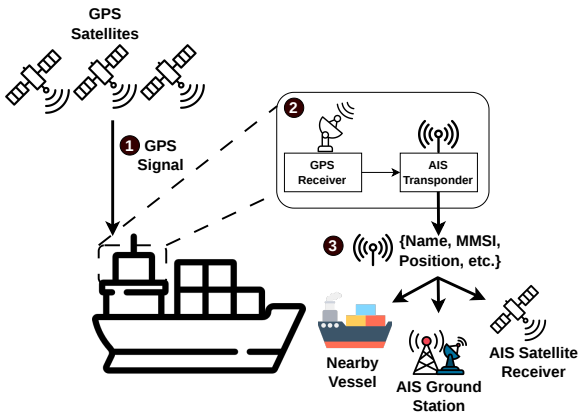


FIGURE 1. End-to-end AIS pipeline: **(1)** GPS satellites transmit positioning signals to the vessel. **(2)** The onboard GPS receiver computes a position fix and passes it to the AIS transponder, which packages it with static vessel information. **(3)** The transponder broadcasts this data over open VHF channels – unauthenticated and unverified – to nearby vessels, shore-based ground stations, and satellite receivers.

At the core of this pipeline is the AIS transponder itself, whose internal architecture is shown in Figure 2. The transponder manages two distinct data flows. On the transmit path, GPS/GNSS positioning data, dynamic sensor inputs such as heading and speed, and manually configured static vessel data are packaged by a message encoder, scheduled for transmission by a TDMA scheduler, and broadcast over VHF. On the receive path, incoming VHF messages from other vessels are decoded by a message parser and forwarded to onboard navigation displays via NMEA 0183 or NMEA 2000 data links. Critically, no authentication is applied at any stage of either path.

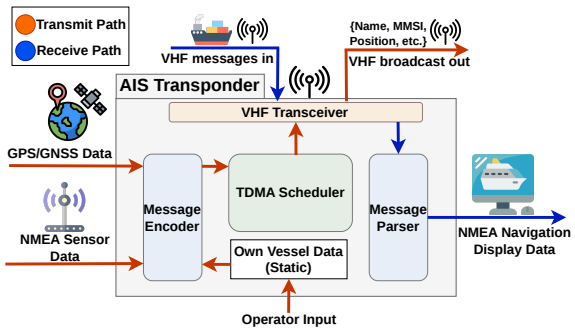


FIGURE 2. Internal architecture of an AIS transponder. Transmit path: GPS/GNSS positioning data, NMEA sensor inputs, and static vessel data are packaged by the Message Encoder, scheduled by the TDMA Scheduler, and broadcast over VHF via the transceiver. Receive path: incoming VHF messages from other vessels are decoded by the Message Parser and passed to navigation display systems via NMEA 0183/2000.

AIS messages are transmitted using a fixed binary format encoded over VHF and wrapped in NMEA 0183 sentences for onboard processing. A standard position report (the most common message type) takes the form `!AIVDM,<count>,<index>,<channel>,<payload>,<fill>*<checksum>`, where the binary payload encodes the vessel's MMSI, latitude and longitude, speed and course over ground, heading, and navigational status. The checksum field validates sentence structure only and is not a cryptographic integrity guarantee over the payload content itself.

The standard defines two primary transponder classes. Class A is mandated for commercial vessels over 300 gross tons and transmits at high power (up to 12.5W) with update intervals as short as two seconds when underway. Class B, intended for smaller and leisure vessels, transmits at lower power (up to 2W) with 30-second intervals and fewer required data fields [9]. Both classes operate on the same open VHF channels with the same absence of authentication. Class A's higher priority and transmission frequency makes its spoofed messages correspondingly more impactful, as they are treated as authoritative by receiving systems.

This AIS system was designed with a clear and narrow purpose: collision avoidance and navigational safety. In this context, open broadcast made sense: the more receivers that could hear a vessel's position, the safer the waterway. Security was not a design consideration. There is no mechanism for a receiver to verify that a broadcast originated from the vessel it claims to represent, that the position data is accurate, or that the message has not been tampered with in transit [10]. Every broadcast is implicitly trusted. This is not a problem that encryption alone can solve. Even if AIS messages were wrapped in TLS or a similar transport-layer security protocol, the fundamental issue would remain: there is no authority to verify that the vessel claiming a given identity or position is telling the truth. AIS security is a distributed trust problem; any vessel can broadcast any data, and the system has no mechanism to adjudicate between legitimate and falsified claims at scale.

This trust assumption has become a critical liability. AIS data flows directly into Electronic Chart Display and Information Systems (ECDIS), voyage planning tools, and traffic monitoring infrastructure – all without independent validation. An attacker who can inject a convincing AIS message, or manipulate the GPS input that feeds a transponder, gains the ability to place any vessel anywhere on the charts of everyone within reception range. The system designed to make the seas safer has, in the absence of any authentication layer, become a vector for precisely the kind of deception it was meant to prevent.

AIS Attack Taxonomy

The vulnerabilities introduced by AIS's open, unauthenticated design manifest across multiple layers of the navigation stack. Attacks can target the radio signals feeding into a transponder, the data a vessel chooses to broadcast, or the software systems that process AIS messages downstream. Each layer presents distinct attack surfaces and requires distinct defensive approaches, as illustrated in Figure 3.

Signal-Layer Attacks. The most direct way to corrupt AIS data is to corrupt the GPS signal feeding it. In an external spoofing attack, an adversary transmits a counterfeit GPS signal stronger than the genuine satellite signal, causing the victim receiver to compute an incorrect position – one the receiver has no way to distinguish from a legitimate fix [6]. Jamming takes a cruder approach, flooding the GPS frequency band with noise to deny positioning entirely. Both attacks are difficult to detect in real time because the receiver behaves normally from its own perspective; it simply has the wrong answer.

Data-Layer Attacks. An attacker with access to an AIS transponder, or the ability to inject raw AIS messages over VHF, can manipulate broadcasts directly without touching GPS at all. Self-spoofing involves a vessel deliberately falsifying its own

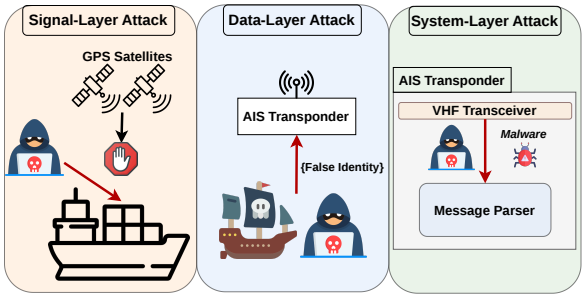


FIGURE 3. AIS attack taxonomy across three layers. **Signal-layer attacks** target the GPS signal before it reaches the vessel, corrupting the position fix fed into the AIS transponder. **Data-layer attacks** manipulate AIS broadcasts directly, injecting false identities or fabricating vessels without touching the underlying GPS signal. **System-layer attacks** exploit software vulnerabilities in the AIS message parser, allowing malformed or malicious messages to corrupt downstream navigation systems.

reported position or identity, a tactic used to evade monitoring or sanctions enforcement [11]. Message injection allows an attacker to fabricate vessels that do not exist, creating so-called ghost ships that appear on traffic displays and can be used to mislead or distract operators. Identity manipulation (cloning a legitimate vessel's MMSI) enables an attacker to impersonate known ships, undermining the reliability of the entire identification system.

System-Layer Attacks. Beyond signal and data manipulation, the software that parses and processes AIS messages introduces its own attack surface. AIS receivers and chart systems have been shown to contain vulnerabilities including insecure parsing routines and insufficient input validation, meaning that a malformed or adversarially crafted message can crash, corrupt, or compromise downstream navigation systems [12]. Malware targeting shipboard IT infrastructure can similarly interfere with AIS processing pipelines, potentially manipulating what crews see without any external signal manipulation at all. Together, these three layers illustrate why AIS security cannot be reduced to any single fix. An authentication mechanism that addresses signal-layer spoofing does nothing to prevent data-layer injection; hardened software does not protect against a jammed GPS receiver.

Spoofing as a Weapon of Modern Conflict

Of the attack classes described above, location spoofing has emerged as the most strategically exploited. Analysis of global AIS data reveals that spoofing clusters persistently in conflict-adjacent regions, including the Black Sea, the Eastern Mediterranean, and the Strait of Hormuz [6]. This geographic concentration is not coincidental. It reflects the deliberate deployment of spoofing as an instrument of modern conflict, where navigation disruption can achieve strategic objectives without a single kinetic action.

The motivations are varied. Spoofing can misdirect commercial vessels away from intended routes, disrupt traffic through critical chokepoints, interfere with the navigation of maritime drones and unmanned systems, and provide plausible deniability: an adversary whose fingerprints appear nowhere in the physical domain can deny responsibility for a grounding or collision. For state actors operating in contested maritime environments, this combination of effectiveness and deniability makes spoofing a particularly attractive tool. Compounding this is the low barrier to entry: software-defined radio hardware capable of GPS spoofing is commercially available for as little as a few hundred to a few thousand dollars [13], meaning the same capability available to a well-resourced state actor is accessible to a non-state actor, a proxy, or even an individual. Few offensive tools in the maritime domain offer this combination of strategic impact and accessibility.

A characteristic signature of large-scale, coordinated spoofing has emerged across these regions: vessels appearing to trace perfect circular paths or linear paths over land and water, an artifact of spoofed coordinates being fed into AIS transponders without any geographic sanity checking. This pattern, first observed persistently in the Black Sea and South China Sea [14], [15], has since appeared in the Strait of Hormuz during recent conflict escalations [7], suggesting a common playbook being applied across geopolitically sensitive waterways. Nowhere is the strategic context sharper than the Strait of Hormuz. The strait is one of the world's most critical maritime chokepoints, with roughly 20% of global oil supply transiting its waters daily [16]. Recent conflict escalation in the region has been accompanied by a marked increase in spoofing activity, degrading navigational awareness for commercial vessels operating in an already congested and high-stakes corridor.

The consequences of spoofing in environments like this are not hypothetical. In May 2025, the MSC



FIGURE 4. (*Top*) The *MSC Antonia* grounded in the Red Sea in May 2025 after a spoofing incident [17]. (*Bottom*) AIS comparison showing the vessel's true route and grounding point (left) versus the spoofed route (right).

Antonia grounded in the Red Sea following a spoofing incident in which the vessel's AIS reported a position inconsistent with its true location [17]. Figure 4 illustrates the divergence between the vessel's actual route and the spoofed track that preceded the grounding. The incident is a concrete illustration of how signal-layer manipulation – invisible to the crew in real time – can translate directly into physical catastrophe.

Open Challenges and Future Research Directions

The incidents described above are symptoms of a domain where offensive capability has outpaced defensive infrastructure. The cascading consequences extend well beyond individual vessels: spoofing-induced disruptions at chokepoints like the Strait of Hormuz ripple outward into global supply chains, insurance markets, and every commercial operator transiting contested waters. Addressing these vulnerabilities is urgent, but the path forward is complicated by a set of deeply entrenched technical and operational challenges. Addressing them will require not only targeted technical solutions but a broader research agenda spanning signal security, data fusion, anomaly detection, and software hardening.

Signal Authentication. Civilian GNSS was designed for accessibility, not integrity. Unlike military GPS variants, civilian signals carry no cryptographic authentication mechanism, meaning a receiver has no way to verify that a signal originated from a legitimate satellite rather than a nearby transmitter. Proposals to address this exist, the European Galileo system's Open Service Navigation Message Authentication (OSNMA) and the GPS Chimera protocol both offer promising approaches [18], [19], but adoption is slow. Additionally, retrofitting existing maritime receivers is costly, and the global installed base of unauthenticated hardware will remain in service for decades. Near-term research opportunities include lightweight authentication schemes compatible with existing receiver hardware, and receiver-side anomaly detection that can flag suspicious signal characteristics without requiring full cryptographic verification.

Validating Unauthenticated AIS Data. Even with authenticated GPS signals, AIS broadcasts themselves remain unverified. No single independent data source (e.g., radar, satellite imagery, historical trajectory models) is sufficient to validate AIS reliably on its own; each carries its own gaps in coverage, latency, and reliability [20]. Doing this at the scale of global maritime traffic compounds the difficulty further. A promising research direction is developing systems that aggregate multiple imperfect sources to flag inconsistencies probabilistically, without requiring any single source to be authoritative or centralized infrastructure to be in place.

Detection Without Ground Truth. Building effective spoofing detectors requires knowing what legitimate navigation behavior looks like. In a domain where spoofing is already widespread and historical AIS data is itself potentially corrupted, clean ground truth is difficult to obtain. Single-vessel anomalies are inherently ambiguous; a position jump could reflect spoofing, a hardware fault, or a software glitch. Multi-vessel correlated anomalies – ships simultaneously reporting identical coordinates, or tracing geometrically implausible paths – provide stronger evidence of external interference [21], but translating these patterns into reliable, low-false-positive detectors at global scale remains an active challenge. Generating synthetic ground truth through simulation, and developing unsupervised detection methods that exploit multi-vessel correlation without labeled data, represent key open research directions.

Legacy and Heterogeneous Systems. Maritime infrastructure evolves slowly. Vessels in service today may carry navigation equipment installed a decade or more ago, sourced from a fragmented ecosystem of vendors with inconsistent security practices and limited update mechanisms. Deploying new authentication protocols or validation systems into this environment requires retrofitting hardware that was never designed to support it, coordinating across a global industry with no centralized update authority, and overcoming economic incentives that favor keeping aging systems in service. On the policy side, research is needed to understand what incentive structures – regulatory mandates, insurance requirements, or port access conditions – are most effective at driving security upgrades across a fragmented global industry. On the technical side, usability-focused work on retrofit solutions that minimize disruption to existing hardware and workflows will be essential to making adoption feasible.

AIS Software Vulnerabilities. Even where modern equipment is deployed, the software processing AIS messages introduces its own risks. Security audits of AIS receivers and chart systems have identified insecure parsing routines, insufficient input validation, and in some cases exploitable memory corruption vulnerabilities [22]. A malformed message crafted to exploit these weaknesses can corrupt or crash downstream navigation systems without any signal-level manipulation. Looking forward, hardening this software layer requires treating the AIS message parser as a security boundary: separating, hardening, and standardizing parser implementations following established secure data intake practices [23] would ensure malformed messages are dropped before reaching downstream systems, prevent mutual misinterpretation between components, and meaningfully reduce the software attack surface. In fact, one of the Accelerators of the DARPA Resilient Software Systems (RSS) program explicitly focuses on hardening the AIS parser [24]. Ensuring that new implementations are developed with security from the outset is a necessary complement to any signal- or data-level solution.

Addressing these challenges in isolation will not be sufficient. Progress will require coordinated investment across academia, industry, and government: shared datasets for detector development, open standards for AIS software security, and international frameworks for incident reporting and response. The research directions outlined above – authenticated signals, multimodal validation, ground-truth generation, and hardened parsers – form a coherent agenda for securing maritime navigation infrastructure. The maritime domain is late to this reckoning, but the cost of further delay is measured in grounded vessels, disrupted supply chains, and compromised national security.

Conclusion

The grounding of the MSC Antonia in the Red Sea was the predictable outcome of a system designed for openness operating in an environment that has become deeply adversarial. AIS was built to make the seas safer by sharing information freely, and for decades, in a world where that information was rarely contested, it did. That world no longer exists.

The Strait of Hormuz has become the sharpest illustration of this shift: a critical artery of global trade, now overlaid with a persistent fog of spoofed positions, manipulated identities, and corrupted navigational data. But the vulnerabilities exposed there are not unique to one region or one conflict. The same unauthenticated signals, the same insecure parsers, and the same implicit trust assumptions exist on every vessel in every waterway on earth.

AIS security is one piece of a much larger problem. Maritime cybersecurity encompasses threats ranging from ransomware targeting port logistics systems to malware on shipboard operational technology – threats that mariners are already encountering and that the industry is underprepared to handle [5]. Navigation integrity is the most immediately safety-critical dimension of this problem, but it cannot be addressed in isolation from the broader effort to harden the cyber-physical systems on which global shipping depends.

The path forward requires treating maritime navigation infrastructure with the same seriousness afforded to other critical systems. That means investing in signal authentication, validating data at scale, hardening software, and building the cross-sector collaboration necessary to make progress on all of these fronts simultaneously. The seas have always been dangerous. The challenge now is ensuring that maritime cybersecurity keeps pace with the threats that have already arrived.

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